

Eigenvalues and inverse problems

A cubic-time Schur algorithm using logarithmic precision

Difficulty: extreme **Importance:** broadly interesting**Status:** Solved

Rating rationale: Extreme reflects an end-to-end precision and complexity theorem for general Schur decomposition; broad impact is justified by this central eigenvalue-computation task.

Resolution — 2026-09-11

Affirmative resolution by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. [Complete manuscript](#) · [PDF](#) · [LaTeX](#). **Theorem 1, Lemmas 2–12 and the final four proof sections.**

The end-to-end algorithm uses $O(n^3 \log^c(n/\delta))$ arithmetic operations and $O(\log(n/\delta))$ mantissa bits, with universal constants, for every complex input with $\|A\|_2 \leq 1$. With probability at least 0.99 it returns an exactly upper triangular T and a Q satisfying both displayed residual bounds. The proof includes finite random sampling, deterministic work caps, global recursive conditioning and floating-point error control, without an input separation or diagonalizability assumption. It is an asymptotic existence result with conservative constants, not a production implementation.

The complete argument received an independent Codex-agent **PASS** on 11 September 2026. The [review report](#) records the exact scope and a hash of the original reviewed manuscript. The mathematical sections remain unchanged in the authored version. Original ChatGPT generation is disclosed; no external human peer review or formal proof certificate is asserted. [Submission and verification record](#).

The original statement, source references and prior audit notes are retained; the former difficulty rating is historical.

Problem statement

Does a randomized floating-point algorithm exist with the following guarantee? Given $A \in \mathbb{C}^{n \times n}$, $\|A\|_2 \leq 1$, and $0 < \delta < 1/2$, it uses at most

$$O(n^3 \log^c(n/\delta)) \text{ arithmetic operations and } O(\log(n/\delta)) \text{ mantissa bits}$$

and, with probability at least 0.99, returns an upper triangular T and Q satisfying

$$\|A - QTQ^*\|_2 \leq \delta, \quad \|Q^*Q - I\|_2 \leq \delta?$$

The constants and exponent c must be universal, with no additional eigenvalue-separation or diagonalizability assumption. Use the usual relative-error floating-point model, with sufficient exponent range to avoid overflow and underflow. This is a statement about the precision needed for a rigorous algorithm, not whether ordinary QR implementations usually work well.

References

Amsel et al., *Linear Systems and Eigenvalue Problems*, Problem 3.3. Schneider, *Pseudospectral Divide-and-Conquer for the Generalized Eigenvalue Problem*, dissertation, §1.6.1 and Chapter 6. Banks et al., *FOCM diagonalization paper*, §6, precision-reduction question.

Earlier status check — 2026-09-08

Searches for *Schur logarithmic precision 2026* and *site:arxiv.org Schur precision 2026* found no complete analysis meeting both bounds. The dissertation explicitly distinguishes analyzed subroutines from an end-to-end precision theorem.

Audit update — 2026-09-10

Rechecked [workshop version 3](#), Problem 3.3 and its update. The report still distinguishes algorithmic ingredients from a complete logarithmic-precision analysis. Targeted cubic-Schur and precision searches found no theorem meeting every displayed requirement.